IAN CAMPBELL

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Technical Skills

Languages: Python, Go, Java, C++, SQL, TypeScript, JavaScript
Backend and Systems: FastAPI, REST APIs, RPC, Concurrency, Distributed Systems, OOP
Frameworks and Data: React, Pandas, NumPy, scikit-learn
Tools and Cloud: Docker, Git, GitHub, GitHub Actions, pytest, AWS S3, CloudFront, Route 53

Projects

Surf Conditions Visualization and Prediction Platform | *Python, FastAPI, React, Pandas* 07/2026 – Present

- React and FastAPI application for processing and visualizing marine conditions for surfers and marine workers.
- Built Python pipelines that ingest realtime NOAA data and standardize timestamps, measurements, and missing values.
- Combining buoy, bathymetry, swell, wind, and tide data into structured inputs for nearshore analysis.
- Automating current condition collection and historical dataset generation for future surf prediction models.

Chess Engine and Full Stack Application | *Python, FastAPI, React, pytest* 05/2026 – 10/2026

- FastAPI backend with REST endpoints for game state, move validation, engine decisions, and a React client.
- Implemented four bot difficulty levels ranging from random move selection through minimax based search.
- Optimized Minimax with Alpha Beta pruning and move ordering, **reducing depth 5 search time from 10ms to 2ms**
- Used pytest to validate API behavior, game state requests and resposnes, bot difficulty, and game evaluation functions.
- Built an opening book based on best lines in chess theory that **reduced the Stockfish evaluation gap from 1.6 to 0.3**.

BU Rent Estimator | *Python, Pandas, scikit-learn, Django, Beautiful Soup* 10/2025 – 12/2025
Boston University: CS506 Applied Machine Learning Capstone Project

- Built a Python pipeline that collected and transformed apartment listings into structured datasets for modeling.
- Processed **1,433 listings** by standardizing fields, handling missing values, and filtering pricing outliers.
- Selected rental features using correlation analysis and trained six neighborhood specific regression models.
- Achieved **R^2 above 0.80 across all six models, with several above 0.90**.

Raft Distributed Consensus | *Go, RPC, Goroutines, Distributed Systems* 10/2025 – 12/2025
Boston University: CS351 Distributed Systems

- Built a Go consensus system that kept replicated logs consistent during server failures and network partitions.
- Implemented leader election and heartbeats using RequestVote, AppendEntries, RPC, and concurrent goroutines.
- Supported log replication, commit delivery, and follower recovery across changing cluster states.
- Tested reliability under server failures, reordered RPCs, disconnects, partitions, and leader recovery.

Education

Boston University 08/2023 – 12/2025
Bachelor of Arts in Computer Science *Boston, MA*

Relevant Coursework: Software Engineering, Full Stack Development, Databases, Data Structures, Algorithms, Distributed Systems, Computer Systems, Data Science Tools & Machine Learning

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Professional Experience

BMW Genius 01/2025 – 10/2026
Asbury Automotive Group, BMW of Boston *Boston, MA*

- Diagnosed BMW software, infotainment, connectivity, mobile app, and configuration issues by finding root causes.
- Built a Python inventory automation tool that **processed 384 vehicles and reduced up to 36 hours of manual work to about five minutes.**